
5. Consider the following database of student enrolment in courses and books adopted each course.

STUDENT (regno: string, name: string, major: string, bdate: date)

COURSE (course: int, cname: string, dept: string)

ENROLL (#regno: string, course#: int, sem:int marks: int)

TEXT (book-ISBN: int, book-title: string, publisher: string, author: string)

BOOK_ADOPTION (course#: int, sem: int, book-ISBN#: int)

1. Create the above tables by properly specifying the primary keys and the foreign Keys.
 - i. Enter at least five tuples for each relation.
 - ii. Demonstrate how you add a textbook to the database and make this book be adapted by some department.
 - iii. Produce list of textbooks (include Course#, Book-ISBN, Book-title) in the alphabetical order for courses offered by the CS department that use more than two books.
 - iv. List any department that has its adopted books published by a specific publisher.

```
CREATE TABLE STUDENT (  
    regno VARCHAR(10) PRIMARY KEY,  
    name VARCHAR(50),  
    major VARCHAR(50),  
    bdate DATE  
);
```

```
CREATE TABLE COURSE (  
    course INT PRIMARY KEY,  
    cname VARCHAR(50),  
    dept VARCHAR(50)  
);
```

```
CREATE TABLE ENROLL (  
    regno VARCHAR(10),  
    course INT,  
    sem INT,  
    marks INT,  
    PRIMARY KEY (regno, course, sem),  
    FOREIGN KEY (regno) REFERENCES STUDENT(regno),  
    FOREIGN KEY (course) REFERENCES COURSE(course)  
);
```

```
CREATE TABLE TEXT (  
    book_ISBN INT PRIMARY KEY,  
    book_title VARCHAR(100),  
    publisher VARCHAR(50),
```

```
author VARCHAR(50)
);
```

```
CREATE TABLE BOOK_ADOPTION (
  course INT,
  sem INT,
  book_ISBN INT,
  PRIMARY KEY (course, sem, book_ISBN),
  FOREIGN KEY (course) REFERENCES COURSE(course),
  FOREIGN KEY (book_ISBN) REFERENCES TEXT(book_ISBN)
);
```

```
INSERT INTO STUDENT VALUES
('S1', 'Amit', 'CS', '2002-01-10'),
('S2', 'Rahul', 'CS', '2001-05-21'),
('S3', 'Sneha', 'EC', '2002-07-15'),
('S4', 'Priya', 'CS', '2001-09-30'),
('S5', 'Kiran', 'ME', '2000-12-11');
```

```
INSERT INTO COURSE VALUES
(101, 'DBMS', 'CS'),
(102, 'OS', 'CS'),
(103, 'Networks', 'EC'),
(104, 'Thermodynamics', 'ME'),
(105, 'Algorithms', 'CS');
```

```
INSERT INTO ENROLL VALUES
('S1', 101, 1, 85),
('S2', 102, 1, 78),
('S3', 103, 1, 88),
('S4', 101, 1, 90),
('S5', 104, 1, 70);
```

```
INSERT INTO TEXT VALUES
(111, 'DBMS Concepts', 'Pearson', 'Korth'),
(112, 'Operating Systems', 'McGrawHill', 'Galvin'),
(113, 'Computer Networks', 'Pearson', 'Tanenbaum'),
(114, 'Thermodynamics Basics', 'Wiley', 'Smith'),
(115, 'Algorithms Design', 'Pearson', 'Cormen');
```

```
INSERT INTO BOOK_ADOPTION VALUES
(101, 1, 111),
```

```
(101, 1, 115),  
(102, 1, 112),  
(103, 1, 113),  
(105, 1, 115);
```

-- ii. Add a new textbook and adopt it for a course

```
INSERT INTO TEXT VALUES (116, 'Advanced DBMS', 'Pearson', 'Navathe');  
INSERT INTO BOOK_ADOPTION VALUES (101, 2, 116);
```

-- iii. List of textbooks for CS courses using > 2 books, ordered by title

```
SELECT C.course, T.book_ISBN, T.book_title  
FROM COURSE C  
JOIN BOOK_ADOPTION B ON C.course = B.course  
JOIN TEXT T ON B.book_ISBN = T.book_ISBN  
WHERE C.dept = 'CS'  
AND C.course IN (  
    SELECT course FROM BOOK_ADOPTION  
    GROUP BY course HAVING COUNT(book_ISBN) > 2  
)  
ORDER BY T.book_title;
```

-- iv. List departments using books from a specific publisher (e.g., 'Pearson')

```
SELECT DISTINCT C.dept  
FROM COURSE C  
JOIN BOOK_ADOPTION B ON C.course = B.course  
JOIN TEXT T ON B.book_ISBN = T.book_ISBN  
WHERE T.publisher = 'Pearson';
```

